ELITE IGCSE ACADEMY

Student Past Paper Solutions

Nov 2023 P2H Worked Solutions

Nov 2023 | P2H

29 questions ▪ question image followed by worked solution

PREPARED BY

Dr. Eslam Ahmed

Assistant Lecturer, Cairo University Faculty of Engineering

WhatsApp: +20 112 000 9622 | eliteigcse.com

STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION

Topic: **Statistics Toolkit**

Nov 2023 P2H - Nov 2023 | P2H

Q#: 1 Marks: 4

1 The table shows information about the lengths, in minutes, of 50 telephone calls.

Length of telephone call (m minutes)	Frequency
$0 < m \leqslant 5$	8
$5 < m \leqslant 10$	2
$10 < m \leqslant 15$	6
$15 < m \leqslant 20$	4
$20 < m \leqslant 25$	12
$25 < m \leqslant 30$	18

(a) Write down the modal class.

....

(1)

(b) Work out an estimate for the total length, in minutes, of these telephone calls.

.... minutes

(3)

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION Q#: 1 Marks: 4

TOPIC: **Statistics Toolkit**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

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Topic check. Statistics Toolkit. The tag is correct.

Solution

The modal class is the class with the greatest frequency:

$$25 < m \leq 30$$

Use midpoints $2.5, 7.5, 12.5, 17.5, 22.5, 27.5$.

$$2.5(8) + 7.5(2) + 12.5(6) + 17.5(4) + 22.5(12) + 27.5(18) = 945$$

FINAL ANSWER

modal class $25 < m \leq 30$, estimated total 945 minutes.

PAST PAPER QUESTION Q#: 2 Marks: 4

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2023 P2H - Nov 2023 | P2H

2 The diagram shows triangle ABC and triangle ECD

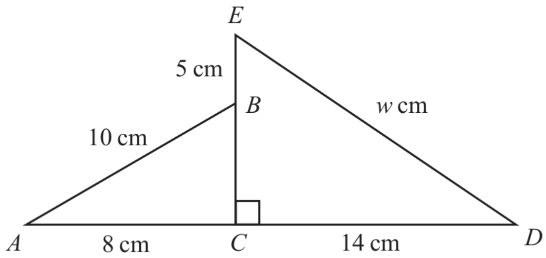

Diagram **NOT** accurately drawn

ACD and EBC are straight lines.

$AB = 10$ cm $\quad AC = 8$ cm $\quad EB = 5$ cm $\quad CD = 14$ cm $\quad ED = w$ cm

Work out the value of w
Give your answer correct to one decimal place.

$w =$ ....

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 4

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

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Topic check. Corrected to right-angled triangles.

Solution

In triangle ABC,

$$BC^2 = 10^2 - 8^2$$

$$BC^2 = 36$$

$$BC = 6$$

So

$$EC = EB + BC = 5 + 6 = 11$$

In triangle ECD,

$$w^2 = 11^2 + 14^2$$

$$w^2 = 317$$

$$w = 17.804\ldots$$

FINAL ANSWER

$w = 17.8$ cm to 1 decimal place.

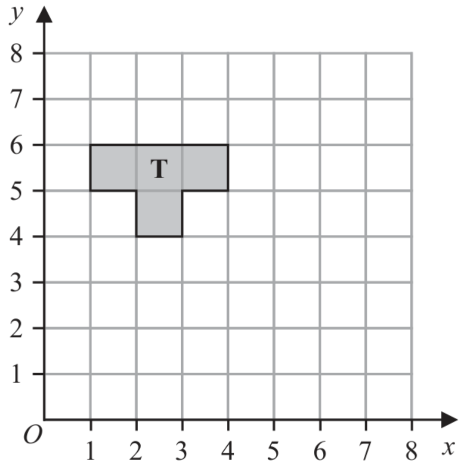
PAST PAPER QUESTION

Q#: 3 Marks: 5

TOPIC: **Transformations**

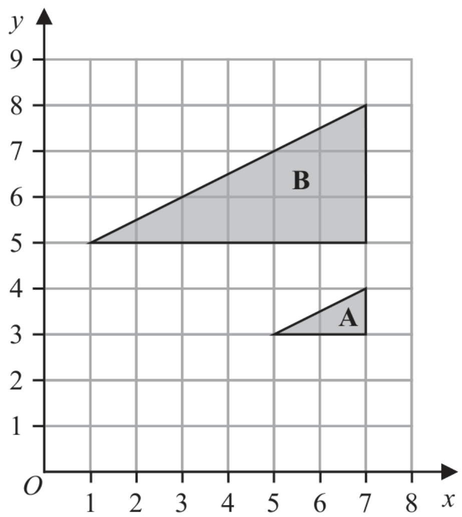
Nov 2023 P2H - Nov 2023 | P2H

3



(a) Reflect shape **T** in the line $y = x$

(2)



(b) Describe fully the single transformation that maps triangle **A** onto triangle **B**

....

....

(3)

(Total for Question 3 is 5 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 5

TOPIC: **Transformations**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

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Topic check. Transformations. The tag is correct.

Solution

(a) Reflection in $y = x$ swaps the coordinates:

$$(x, y) \mapsto (y, x)$$

The reflected T-shape has vertices

$$(6, 1), (6, 4), (5, 4), (5, 3), (4, 3), (4, 2), (5, 2), (5, 1)$$

(b) Triangle A is mapped to B by an enlargement with scale factor 3 and centre $(7, 2)$.

FINAL ANSWER

(a) reflected vertices $(6, 1), (6, 4), (5, 4), (5, 3), (4, 3), (4, 2), (5, 2), (5, 1)$; (b) enlargement, scale factor 3, centre $(7, 2)$.

PAST PAPER QUESTION

Q#: 4 Marks: 8

TOPIC: **Algebraic Roots & Indices**

Nov 2023 P2H - Nov 2023 | P2H

4 (a) Solve $\dfrac{2x+5}{6} = 2x - 5$

Show clear algebraic working.

$x =$ ....

(3)

(b) Simplify $h^{15} \div h^{3}$

....

(1)

(c) Simplify fully $\left(2g^{3}k^{5}\right)^{4}$

....

(2)

(d) Given that $\dfrac{y^{5} \times y^{n}}{y^{7}} = y^{12}$

work out the value of n

$n =$ ....

(2)

(Total for Question 4 is 8 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 8

TOPIC: **Algebraic Roots & Indices**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

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Topic check. Corrected to algebraic roots and indices. Index laws carry most of the marks.

Solution

(a)

$$\frac{2x+5}{6} = 2x - 5$$

$$2x + 5 = 12x - 30$$

$$35 = 10x$$

$$x = \frac{7}{2}$$

(b)

$$h^{15} \div h^3 = h^{15-3} = h^{12}$$

(c)

$$(2g^3k^5)^4 = 2^4g^{12}k^{20} = 16g^{12}k^{20}$$

(d)

$$\frac{y^5 \times y^n}{y^7} = y^{5+n-7} = y^{n-2}$$

Given that this equals y^{12},

$$n - 2 = 12$$

$$n = 14$$

FINAL ANSWER

$x = \frac{7}{2}$, h^{12}, $16g^{12}k^{20}$, and $n = 14$.

PAST PAPER QUESTION

TOPIC: **Ratio Toolkit**

Nov 2023 P2H - Nov 2023 | P2H

Q#: 5 Marks: 3

5 Avril bakes a cake.

She uses flour, butter and sugar such that

weight of flour : weight of butter = 6 : 5
weight of butter : weight of sugar = 3 : 2

Avril uses 120 grams of sugar.

Work out the weight of flour Avril uses.

.... grams

(Total for Question 5 is 3 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 3

TOPIC: **Ratio Toolkit**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Ratio Toolkit. The tag is correct.

Method

Butter : sugar $= 3 : 2$.
Sugar is 120 g, so

$$2 \text{ parts} = 120$$

$$1 \text{ part} = 60$$

$$\text{butter} = 3 \times 60 = 180$$

Flour : butter $= 6 : 5$.

$$5 \text{ parts} = 180$$

$$1 \text{ part} = 36$$

$$\text{flour} = 6 \times 36 = 216$$

FINAL ANSWER

216 grams

PAST PAPER QUESTION Q#: 6 Marks: 3

TOPIC: **Algebraic Proof**

Nov 2023 P2H - Nov 2023 | P2H

6 Show that $3\frac{3}{7} \div 2\frac{2}{3} = 1\frac{2}{7}$

(Total for Question 6 is 3 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 3

TOPIC: **Algebraic Proof**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

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Topic check. Algebraic proof. The tag is acceptable because the question asks to show the exact result.

Solution

$$3\frac{3}{7} = \frac{24}{7}, \qquad 2\frac{2}{3} = \frac{8}{3}$$

$$3\frac{3}{7} \div 2\frac{2}{3} = \frac{24}{7} \div \frac{8}{3}$$

$$= \frac{24}{7} \times \frac{3}{8} = \frac{9}{7} = 1\frac{2}{7}$$

FINAL ANSWER

$1\frac{2}{7}$.

PAST PAPER QUESTION Q#: 7 Marks: 3

TOPIC: **Compound Interest & Depreciation**

Nov 2023 P2H - Nov 2023 | P2H

7 Hermione buys a boat for $26 800
The value of the boat depreciates by 8% each year.

Work out the value of the boat at the end of 3 years.
Give your answer correct to the nearest dollar.

$..

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 3

TOPIC: **Compound Interest & Depreciation**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

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Topic check. Compound Interest and Depreciation. The tag is correct.

Method

A depreciation of 8% gives multiplier

$$0.92$$

After 3 years,

$$26800(0.92)^3 = 20868.8384$$

Correct to the nearest dollar,

$$20868.8384 \approx 20869$$

FINAL ANSWER

$20869

PAST PAPER QUESTION

Q#: 8 Marks: 3

TOPIC: **Statistics Toolkit**

Nov 2023 P2H - Nov 2023 | P2H

8 The mean number of goals scored by a hockey team in 8 matches is 6
The team plays 2 more matches and scores k goals in each match.
The mean number of goals scored by the hockey team in the 10 matches is 7

Work out the value of k

k = ...

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 3

TOPIC: **Statistics Toolkit**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

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Topic check. Statistics Toolkit. The tag is correct.

Solution

The total for the first 8 matches is

$$8 \times 6 = 48$$

The total for all 10 matches is

$$10 \times 7 = 70$$

So the last two matches have total

$$70 - 48 = 22$$

The last two scores are k and $k + 10$:

$$k + (k + 10) = 22$$

$$2k + 10 = 22$$

$$k = 6$$

FINAL ANSWER

$k = 6$.

PAST PAPER QUESTION Q#: 9 Marks: 2

TOPIC: **Linear Graphs (y = mx + c)**

Nov 2023 P2H - Nov 2023 | P2H

9 A straight line passes through the points with coordinates (0, –3) and (2, 0)

Find an equation of the line.

....

(Total for Question 9 is 2 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 2

TOPIC: **Linear Graphs (y = mx + c)**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

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Topic check. Corrected to linear graphs. The question asks for an equation of a straight line through two points.

Solution

The line passes through $(0, -3)$ and $(2, 0)$.
The gradient is

$$\frac{0-(-3)}{2-0} = \frac{3}{2}$$

The y-intercept is -3, so

$$y = \frac{3}{2}x - 3$$

FINAL ANSWER

$y = \frac{3}{2}x - 3$.

PAST PAPER QUESTION

Q#: 10 Marks: 5

TOPIC: **Area & Perimeter**

Nov 2023 P2H - Nov 2023 | P2H

10 The diagram shows a hexagon *ABCDEF*

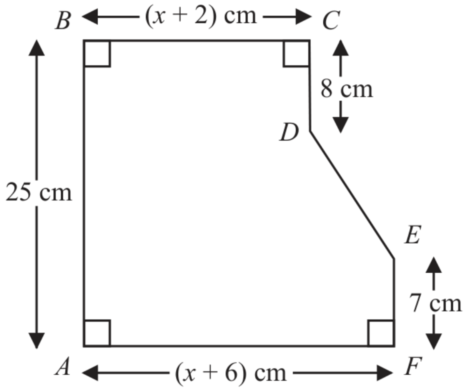

Diagram **NOT** accurately drawn

$AB = 25$ cm $BC = (x + 2)$ cm $CD = 8$ cm $EF = 7$ cm $AF = (x + 6)$ cm

The area of hexagon *ABCDEF* is 258 cm²

Work out the value of x

$x =$ ....

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 5

TOPIC: **Area & Perimeter**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

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Topic check. Area & Perimeter. The tag is correct.

Solution

The area of the hexagon can be found from coordinates or by splitting the shape.
It simplifies to

$$\text{area} = 25x + 98$$

Given the area is 258 cm^2,

$$25x + 98 = 258$$

$$25x = 160$$

$$x = 6.4$$

FINAL ANSWER

$x = 6.4$.

PAST PAPER QUESTION

Q#: 11 Marks: 5

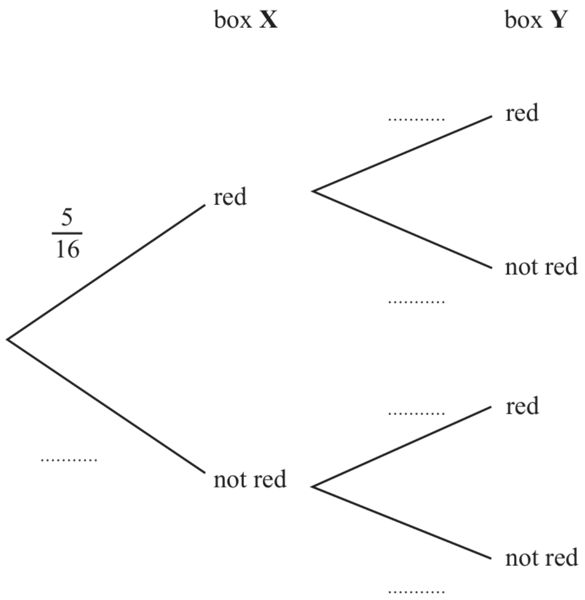
TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Nov 2023 P2H - Nov 2023 | P2H

11 Sid has 2 boxes of crayons, box **X** and box **Y**

5 of the 16 crayons in box **X** are red.
7 of the 20 crayons in box **Y** are red.
Sid takes at random one crayon from box **X** and one crayon from box **Y**

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Sid takes two crayons that are red or two crayons that are not red.

....

(3)

(Total for Question 11 is 5 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 5

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

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Topic check. Probability Diagrams - Venn & Tree Diagrams. The tag is correct.

Solution

Box X has 5 red and 11 not red crayons.
Box Y has 7 red and 13 not red crayons.
Same colour type means both red or both not red.

$$P(\text{same}) = \frac{5}{16} \times \frac{7}{20} + \frac{11}{16} \times \frac{13}{20}$$
$$= \frac{35}{320} + \frac{143}{320}$$
$$= \frac{89}{160}$$

FINAL ANSWER

$\frac{89}{160}$.

PAST PAPER QUESTION Q#: 12 Marks: 4

TOPIC: **Algebraic Roots & Indices**

Nov 2023 P2H - Nov 2023 | P2H

12 $2^7 \times 4^5 = 4^x$

(a) Calculate the value of x

$x =$ ....

(2)

(b) Simplify fully $\left(125p^6y^{24}\right)^{\frac{2}{3}}$

....

(2)

(Total for Question 12 is 4 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 4

TOPIC: **Algebraic Roots & Indices**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

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Topic check. Corrected to algebraic roots and indices. Both parts are index-law questions.

Solution

(a)

$$2^7 \times 4^5 = 2^7 \times (2^2)^5 = 2^{17}$$

Also

$$4^x = (2^2)^x = 2^{2x}$$

So

$$2x = 17$$

$$x = \frac{17}{2}$$

(b)

$$(125p^6y^{24})^{2/3} = 125^{2/3}p^4y^{16}$$

$$= 25p^4y^{16}$$

FINAL ANSWER

(a) $x = \frac{17}{2}$, (b) $25p^4y^{16}$

PAST PAPER QUESTION

TOPIC: **Statistics Toolkit**

Nov 2023 P2H - Nov 2023 | P2H

Q#: 13 Marks: 2

13 Robert asked 11 people how many meetings they attended last week.

Here are the results in numerical order.

1	2	4	6	6	8	11	12	13	14	17

Find the interquartile range of the number of meetings.

....

(Total for Question 13 is 2 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 2

TOPIC: **Statistics Toolkit**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Statistics Toolkit. The tag is correct.

Solution

The results in order are

$$1,\ 2,\ 4,\ 6,\ 6,\ 8,\ 11,\ 12,\ 13,\ 14,\ 17$$

There are 11 values.

$$Q_1 = 4, \qquad Q_3 = 13$$

$$\text{IQR} = 13 - 4 = 9$$

FINAL ANSWER

9.

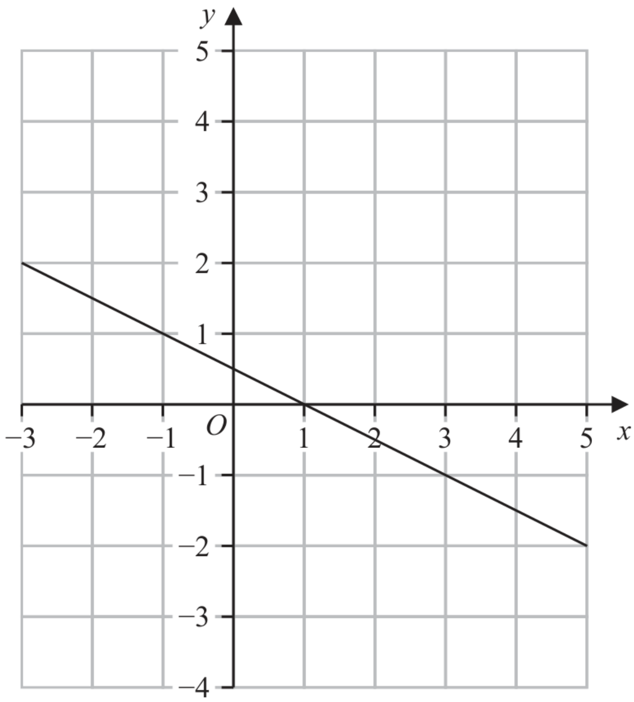
PAST PAPER QUESTION

TOPIC: **Simultaneous Equations**

Nov 2023 P2H - Nov 2023 | P2H

Q#: 14 Marks: 3

14 Here is the graph of the equation $2y + x = 1$ drawn on a grid.



By drawing another straight line on the grid, solve the simultaneous equations

$$y - x - 2 = 0$$
$$2y + x = 1$$

$x =$ ....

$y =$ ....

(Total for Question 14 is 3 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 3

TOPIC: **Simultaneous Equations**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Simultaneous equations. The tag is correct.

Solution

The equations are

$$y - x - 2 = 0$$

$$2y + x = 1$$

From the first equation,

$$y = x + 2$$

Substitute into the second equation:

$$2(x + 2) + x = 1$$

$$3x + 4 = 1$$

$$3x = -3$$

$$x = -1$$

Then

$$y = -1 + 2 = 1$$

FINAL ANSWER

$x = -1$, $y = 1$.

PAST PAPER QUESTION

Q#: 15 **Marks: 5**

TOPIC: **Algebraic Proof**

Nov 2023 P2H - Nov 2023 | P2H

15 (a) Use algebra to show that $0.3\dot{7}\dot{2} = \dfrac{41}{110}$

(2)

(b) Express $\dfrac{\sqrt{125}+\sqrt{80}}{\sqrt{3}}$ in the form $\sqrt{n}$ where n is an integer.

Show your working clearly.

....

(3)

(Total for Question 15 is 5 marks)

PAST PAPER SOLUTION Q#: 15 Marks: 5

TOPIC: **Algebraic Proof**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Algebraic proof. The tag is correct.

Solution

(a) Let

$$x = 0.3727272\ldots$$

Then

$$10x = 3.727272\ldots$$

and

$$1000x = 372.727272\ldots$$

Subtract:

$$990x = 369$$

$$x = \frac{369}{990} = \frac{41}{110}$$

(b)

$$\frac{\sqrt{125}+\sqrt{80}}{\sqrt{3}} = \frac{5\sqrt{5}+4\sqrt{5}}{\sqrt{3}}$$

$$= \frac{9\sqrt{5}}{\sqrt{3}}$$

Rationalise:

$$\frac{9\sqrt{5}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{9\sqrt{15}}{3} = 3\sqrt{15}$$

$$3\sqrt{15} = \sqrt{135}$$

FINAL ANSWER

$0.3727272\ldots = \frac{41}{110}$, and $n = 135$.

PAST PAPER QUESTION Q#: 16 Marks: 3

TOPIC: **Expanding Brackets**

Nov 2023 P2H - Nov 2023 | P2H

16 Expand and simplify $(2x + 3)(x - 5)(x + 4)$

....

(Total for Question 16 is 3 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 3

TOPIC: **Expanding Brackets**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Expanding brackets. The tag is correct.

Solution

$$(x-5)(x+4) = x^2 - x - 20$$

$$(2x+3)(x^2 - x - 20) = 2x^3 + x^2 - 43x - 60$$

FINAL ANSWER

$2x^3 + x^2 - 43x - 60$

PAST PAPER QUESTION Q#: 17 Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Nov 2023 P2H - Nov 2023 | P2H

17 $P = a(c + y)$

$a =$ 8.3 correct to 2 significant figures
$c =$ 2 correct to 1 significant figure
$y =$ 15 correct to the nearest 5

Work out the upper bound for the value of P
Show your working clearly.

....

(Total for Question 17 is 3 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Rounding, Estimation and Bounds. The tag is correct.

Method

$$P = a(c + y)$$

For the upper bound, use the upper bounds of all three values.

$$a < 8.35, \quad c < 2.5, \quad y < 17.5$$

$$P_{\text{upper}} = 8.35(2.5 + 17.5) = 167$$

FINAL ANSWER

167

PAST PAPER QUESTION

Q#: 18 Marks: 4

TOPIC: **Differentiation**

Nov 2023 P2H - Nov 2023 | P2H

18 A particle is moving along a straight line that passes through the fixed point O
The displacement, s metres, of the particle from O at time t seconds is given by

$$s = 2t^3 - 5t^2 + 6t - 5$$

Find the value of t when the acceleration of the particle is 5 m/s²

t = ...

(Total for Question 18 is 4 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 4

TOPIC: **Differentiation**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

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Topic check. Corrected to Differentiation. Acceleration is found from the second derivative of displacement.

Method

$$s = 2t^3 - 5t^2 + 6t - 5$$

Velocity is

$$\frac{ds}{dt} = 6t^2 - 10t + 6$$

Acceleration is

$$\frac{d^2s}{dt^2} = 12t - 10$$

The acceleration is 5, so

$$12t - 10 = 5$$

$$12t = 15$$

$$t = \frac{15}{12} = \frac{5}{4}$$

FINAL ANSWER

$t = \frac{5}{4}$ seconds

PAST PAPER QUESTION

TOPIC: **Functions**

Nov 2023 P2H - Nov 2023 | P2H

Q#: 19 Marks: 7

19 The functions f and g are such that

$$f : x \mapsto 5x + 7$$

$$g : x \mapsto \frac{5}{2x - 9}$$

(a) State which value of x cannot be included in any domain of g

....

(1)

(b) Find fg(4)

....

(2)

The function h is such that

$$h : x \mapsto 3x^2 - 12x + 8 \quad \text{where } x > 2$$

(c) Express the inverse function h^{-1} in the form $h^{-1} : x \mapsto \ldots$

$h^{-1} : x \mapsto$ ....

(4)

(Total for Question 19 is 7 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 7

TOPIC: **Functions**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

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Topic check. Functions. The tag is correct.

Solution

$$f(x) = 5x + 7, \qquad g(x) = \frac{5}{2x - 9}$$

(a) The denominator of g cannot be zero:

$$2x - 9 = 0$$

$$x = \frac{9}{2}$$

(b)

$$g(4) = \frac{5}{8 - 9} = -5$$

$$fg(4) = f(-5) = 5(-5) + 7 = -18$$

(c)

$$h(x) = 3x^2 - 12x + 8, \qquad x > 2$$

Complete the square:

$$h(x) = 3(x - 2)^2 - 4$$

Let

$$y = 3(x - 2)^2 - 4$$

$$y + 4 = 3(x - 2)^2$$

$$x - 2 = \sqrt{\frac{y + 4}{3}}$$

Since $x > 2$, take the positive root:

$$x = 2 + \sqrt{\frac{y + 4}{3}}$$

So

$$h^{-1}(x) = 2 + \sqrt{\frac{x + 4}{3}}$$

FINAL ANSWER

Excluded value $\frac{9}{2}$, $fg(4) = -18$, and $h^{-1}(x) = 2 + \sqrt{\frac{x+4}{3}}$.

PAST PAPER QUESTION Q#: 20 Marks: 4

TOPIC: **Circles, Arcs & Sectors**

Nov 2023 P2H - Nov 2023 | P2H

20 The diagram shows equilateral triangle *ABC* with sides of length 10 cm.
A circle is drawn inside the triangle.

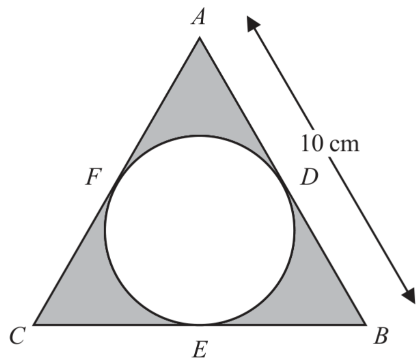

Diagram **NOT** accurately drawn

D, *E* and *F* are points on the circle.

ADB, *BEC* and *CFA* are tangents to the circle.

Calculate the total area of the regions shown shaded in the diagram.
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Circles, Arcs & Sectors**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

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Topic check. Moved to Circles, Arcs & Sectors.

Method

The triangle is equilateral with side length 10 cm.
Area of the triangle:

$$\frac{\sqrt{3}}{4}(10)^2 = 25\sqrt{3}$$

For an equilateral triangle, the inradius is

$$r = \frac{s\sqrt{3}}{6}$$

So

$$r = \frac{10\sqrt{3}}{6} = \frac{5\sqrt{3}}{3}$$

Area of the circle:

$$\pi r^2 = \pi\left(\frac{5\sqrt{3}}{3}\right)^2 = \frac{25\pi}{3}$$

The shaded area is

$$25\sqrt{3} - \frac{25\pi}{3} = 17.1213\ldots$$

FINAL ANSWER

$\boxed{17.1 \text{ cm}^2}$

PAST PAPER QUESTION Q#: 20 Marks: 4

TOPIC: **Circles, Arcs & Sectors**

Nov 2023 P2H - Nov 2023 | P2H

20 The diagram shows equilateral triangle *ABC* with sides of length 10 cm.
A circle is drawn inside the triangle.

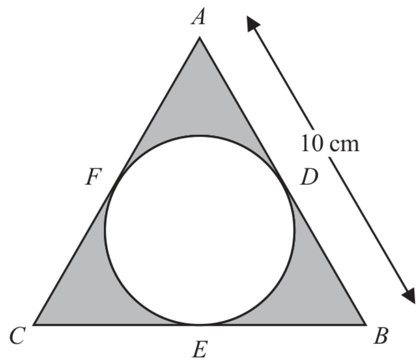

Diagram **NOT** accurately drawn

D, *E* and *F* are points on the circle.

ADB, *BEC* and *CFA* are tangents to the circle.

Calculate the total area of the regions shown shaded in the diagram.
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION Q#: 20 Marks: 4

TOPIC: **Circles, Arcs & Sectors**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved to Circles, Arcs & Sectors.

Method

The triangle is equilateral with side length 10 cm.
Area of the triangle:

$$\frac{\sqrt{3}}{4}(10)^2 = 25\sqrt{3}$$

For an equilateral triangle, the inradius is

$$r = \frac{s\sqrt{3}}{6}$$

So

$$r = \frac{10\sqrt{3}}{6} = \frac{5\sqrt{3}}{3}$$

Area of the circle:

$$\pi r^2 = \pi\left(\frac{5\sqrt{3}}{3}\right)^2 = \frac{25\pi}{3}$$

The shaded area is

$$25\sqrt{3} - \frac{25\pi}{3} = 17.1213\ldots$$

FINAL ANSWER

$\boxed{17.1\ \text{cm}^2}$

PAST PAPER QUESTION

Q#: 21 Marks: 5

TOPIC: **Coordinate Geometry**

Nov 2023 P2H - Nov 2023 | P2H

21 The line with equation $x + 2y = 5$ intersects the curve with equation $x^2 + 3y^2 = 13$ at the points A and B

Find the coordinates of A and the coordinates of B
Show clear algebraic working.

(.............................. , ...)

(.............................. , ...)

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION Q#: 21 Marks: 5

TOPIC: **Coordinate Geometry**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The line is

$$x + 2y = 5$$

So

$$x = 5 - 2y$$

Substitute into

$$x^2 + 3y^2 = 13$$

$$(5 - 2y)^2 + 3y^2 = 13$$

$$25 - 20y + 4y^2 + 3y^2 = 13$$

$$7y^2 - 20y + 12 = 0$$

Factorise:

$$(7y - 6)(y - 2) = 0$$

So

$$y = \frac{6}{7} \quad \text{or} \quad y = 2$$

Using $x = 5 - 2y$:

$$y = \frac{6}{7} \Rightarrow x = \frac{23}{7}$$

$$y = 2 \Rightarrow x = 1$$

FINAL ANSWER

$\left(\frac{23}{7}, \frac{6}{7}\right)$ and $(1, 2)$

PAST PAPER QUESTION Q#: 21 Marks: 5

TOPIC: **Coordinate Geometry**

Nov 2023 P2H - Nov 2023 | P2H

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Find the coordinates of A and the coordinates of B
Show clear algebraic working.

(.............................. , ...)

(.............................. , ...)

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 5

TOPIC: **Coordinate Geometry**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

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The line is

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FINAL ANSWER

$\left(\frac{23}{7}, \frac{6}{7}\right)$ and $(1, 2)$

PAST PAPER QUESTION

Q#: 22 Marks: 5

Topic: **Vectors**

Nov 2023 P2H - Nov 2023 | P2H

22

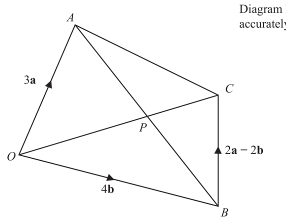


Diagram **NOT** accurately drawn

OACB is a quadrilateral.

$\overrightarrow{OA} = 3\mathbf{a}$ $\overrightarrow{OB} = 4\mathbf{b}$ $\overrightarrow{BC} = 2\mathbf{a} - 2\mathbf{b}$

(a) (i) Find the vector $\overrightarrow{OC}$ in terms of **a** and **b**
Simplify your answer.

$\overrightarrow{OC}$ = ...
(1)

(ii) Find the vector $\overrightarrow{AB}$ in terms of **a** and **b**

$\overrightarrow{AB}$ = ...
(1)

The point *P* lies on *AB* and on *OC*

(b) Using a vector method, find the ratio *AP* : *PB*
Show your working clearly.

....
(3)

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION Q#: 22 Marks: 5

TOPIC: **Vectors**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Ratio Toolkit to Vectors.

Method

$$\overrightarrow{OA} = 3a, \quad \overrightarrow{OB} = 4b, \quad \overrightarrow{BC} = 2a - 2b$$

Part (a):

$$\overrightarrow{OC} = \overrightarrow{OB} + \overrightarrow{BC} = 4b + (2a - 2b) = 2a + 2b$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 4b - 3a$$

For part (b), let P divide AB so that

$$\overrightarrow{AP} = \mu\overrightarrow{AB}$$

Then

$$\overrightarrow{OP} = 3a + \mu(4b - 3a) = (3 - 3\mu)a + 4\mu b$$

Since P lies on OC, also let

$$\overrightarrow{OP} = \lambda(2a + 2b)$$

Equating coefficients:

$$3 - 3\mu = 2\lambda, \quad 4\mu = 2\lambda$$

From $4\mu = 2\lambda$, $\lambda = 2\mu$.

$$3 - 3\mu = 4\mu$$

$$\mu = \frac{3}{7}$$

So $AP : PB = \frac{3}{7} : \frac{4}{7}$.

FINAL ANSWER
$\overrightarrow{OC} = 2a + 2b$, $\overrightarrow{AB} = 4b - 3a$, and $AP : PB = 3 : 4$

PAST PAPER QUESTION

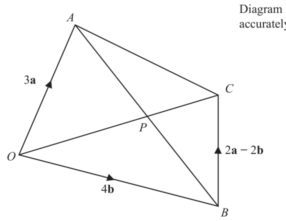
Q#: 22 Marks: 5

TOPIC: **Vectors**

Nov 2023 P2H - Nov 2023 | P2H

22

Diagram **NOT** accurately drawn



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Simplify your answer.

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(1)

(ii) Find the vector $\overrightarrow{AB}$ in terms of **a** and **b**

$\overrightarrow{AB}$ = ...
(1)

The point *P* lies on *AB* and on *OC*

(b) Using a vector method, find the ratio $AP : PB$
Show your working clearly.

....
(3)

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 **Marks: 5**

TOPIC: **Vectors**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Ratio Toolkit to Vectors.

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$$\overrightarrow{OA} = 3a, \quad \overrightarrow{OB} = 4b, \quad \overrightarrow{BC} = 2a - 2b$$

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From $4\mu = 2\lambda$, $\lambda = 2\mu$.

$$3 - 3\mu = 4\mu$$

$$\mu = \frac{3}{7}$$

So $AP : PB = \frac{3}{7} : \frac{4}{7}$.

FINAL ANSWER
$\overrightarrow{OC} = 2a + 2b$, $\overrightarrow{AB} = 4b - 3a$, and $AP : PB = 3 : 4$

PAST PAPER QUESTION Q#: 23 Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

Nov 2023 P2H - Nov 2023 | P2H

23 Here is a frustum of a cone.
The frustum is made by removing a small cone from a similar large cone.

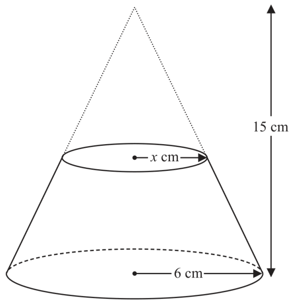


Diagram **NOT** accurately drawn

The height of the large cone is 15 cm.

The radius of the base of the large cone is 6 cm.
The radius of the base of the small cone is x cm.

Given that the volume of the frustum is $\frac{4212}{25}\pi$ cm^3

work out the value of x
Show clear algebraic working.

$x =$ ....

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION Q#: 23 Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The large cone has height 15 cm and radius 6 cm.
The small cone has radius x. Since the cones are similar,

$$\frac{\text{radius}}{\text{height}} = \frac{6}{15} = \frac{2}{5}$$

So for the small cone,

$$\frac{x}{h} = \frac{2}{5}$$

$$h = \frac{5}{2}x$$

Volume of the large cone:

$$\frac{1}{3}\pi(6)^2(15) = 180\pi$$

Volume of the small cone:

$$\frac{1}{3}\pi x^2\left(\frac{5}{2}x\right) = \frac{5}{6}\pi x^3$$

The frustum volume is

$$\frac{4212}{25}\pi$$

So

$$180\pi - \frac{5}{6}\pi x^3 = \frac{4212}{25}\pi$$

Cancel π:

$$180 - \frac{5}{6}x^3 = \frac{4212}{25}$$

$$\frac{5}{6}x^3 = 180 - \frac{4212}{25}$$

$$\frac{5}{6}x^3 = \frac{288}{25}$$

$$x^3 = \frac{1728}{125}$$

$$x = \frac{12}{5} = 2.4$$

FINAL ANSWER

$\boxed{2.4 \text{ cm}}$

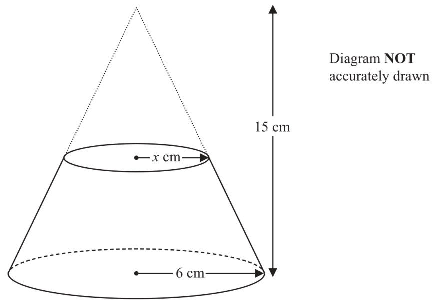
PAST PAPER QUESTION

Q#: 23 Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

Nov 2023 P2H - Nov 2023 | P2H

23 Here is a frustum of a cone.
The frustum is made by removing a small cone from a similar large cone.



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Given that the volume of the frustum is $\frac{4212}{25}\pi$ cm^3

work out the value of x
Show clear algebraic working.

$x =$ ....

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

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The small cone has radius x. Since the cones are similar,

$$\frac{\text{radius}}{\text{height}} = \frac{6}{15} = \frac{2}{5}$$

So for the small cone,

$$\frac{x}{h} = \frac{2}{5}$$

$$h = \frac{5}{2}x$$

Volume of the large cone:

$$\frac{1}{3}\pi(6)^2(15) = 180\pi$$

Volume of the small cone:

$$\frac{1}{3}\pi x^2\left(\frac{5}{2}x\right) = \frac{5}{6}\pi x^3$$

The frustum volume is

$$\frac{4212}{25}\pi$$

So

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Cancel π:

$$180 - \frac{5}{6}x^3 = \frac{4212}{25}$$

$$\frac{5}{6}x^3 = 180 - \frac{4212}{25}$$

$$\frac{5}{6}x^3 = \frac{288}{25}$$

$$x^3 = \frac{1728}{125}$$

$$x = \frac{12}{5} = 2.4$$

FINAL ANSWER

$\boxed{2.4 \text{ cm}}$

PAST PAPER QUESTION

Q#: 24 Marks: 5

TOPIC: **Algebraic Fractions**

Nov 2023 P2H - Nov 2023 | P2H

24 Solve $\dfrac{45x^3 - 80x}{3x^2 + x - 4} \times \left(\dfrac{1}{3x - 4} + \dfrac{1}{x} \right) = \dfrac{4(x+2)}{5x-8}$

Show clear algebraic working.

$x =$ ....

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 5

TOPIC: **Algebraic Fractions**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Solving Quadratic Equations to Algebraic Fractions.

Method

$$\frac{45x^3 - 80x}{3x^2 + x - 4}\left(\frac{1}{3x-4} + \frac{1}{x}\right) = \frac{4(x+2)}{5x-8}$$

Factorise:

$$45x^3 - 80x = 5x(3x-4)(3x+4)$$

$$3x^2 + x - 4 = (3x+4)(x-1)$$

Also,

$$\frac{1}{3x-4} + \frac{1}{x} = \frac{x + 3x - 4}{x(3x-4)} = \frac{4(x-1)}{x(3x-4)}$$

So the left side becomes

$$\frac{5x(3x-4)(3x+4)}{(3x+4)(x-1)} \times \frac{4(x-1)}{x(3x-4)} = 20$$

Therefore,

$$20 = \frac{4(x+2)}{5x-8}$$

$$20(5x-8) = 4(x+2)$$

$$100x - 160 = 4x + 8$$

$$96x = 168$$

$$x = \frac{7}{4}$$

FINAL ANSWER

$x = \frac{7}{4}$

PAST PAPER QUESTION

Q#: 24 Marks: 5

TOPIC: **Algebraic Fractions**

Nov 2023 P2H - Nov 2023 | P2H

24 Solve $\dfrac{45x^3 - 80x}{3x^2 + x - 4} \times \left(\dfrac{1}{3x - 4} + \dfrac{1}{x}\right) = \dfrac{4(x + 2)}{5x - 8}$

Show clear algebraic working.

$x =$ ....

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 5

TOPIC: **Algebraic Fractions**

Nov 2023 P2H - Nov 2023 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Solving Quadratic Equations to Algebraic Fractions.

Method

$$\frac{45x^3 - 80x}{3x^2 + x - 4}\left(\frac{1}{3x-4} + \frac{1}{x}\right) = \frac{4(x+2)}{5x-8}$$

Factorise:

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FINAL ANSWER

$x = \frac{7}{4}$